

fMRI Foundation Model (V2) — Results

1. Pretraining corpus

Five sources, **4627 scans**, harmonized to TR = 0.72 s with a fixed T = 270 window.

Dataset	Native TR (s)	Probe holdout (30%)	Per-batch quota
HCP	0.72	yes	4
ABIDE	per-site	yes	4
OASIS-3	2.2	yes	4
ADNI	3.0	yes	3
AOMIC	0.75/2.0	no (kept whole)	1

Quota = per-batch proportion (proportional sampler). Holdout: 30% of subjects excluded from pretraining -> leakage-free test set.

2. Datasets & scan counts

Every dataset we have, with its role (pretraining / downstream probe) and scan count.

Dataset	Role	Scans	Note
HCP (rest)	pretrain + probe	1084	Sex / Age
ABIDE	pretrain + probe	1035	Autism / Age / Sex
OASIS-3	pretrain + probe	1197	AD Conversion (labels pending)
ADNI	pretrain + probe	812	NC-MCI / AD-HC / Amyloid
AOMIC	pretrain only	~499	derived (4627 - others); no probe
ADHD-200	probe only (external)	162	115 with usable DX label
COBRE	probe only (external)	146	schizophrenia (72 SZ / 74 HC)
UCLA (ds000030)	probe only (external)	265	downloader ready, not yet run
HCP task-fMRI	probe only	—	7 tasks x ~1050, download pending

Pretraining corpus total = 4627 (HCP + ABIDE + OASIS + ADNI + AOMIC). External datasets (ADHD-200 / COBRE / UCLA) were never seen in pretraining.

3. Pretraining ablations (5 SSL runs) — test AUROC

Each run starts from the same DINOv2 (ImageNet) init and changes **one** factor:

- **base**: reference (freeze blocks 0-8) · **fourier**: Fourier positional encoding
- **noblock2**: drop block_2 · **pool**: AvgPool downsampling (instead of strided conv)
- **unfrozen**: all layers unfrozen during SSL

Axis	base	fourier	noblock2	pool	unfrozen
ABIDE / Autism	0.605	0.518	0.484	0.606	0.541
ABIDE / Age	0.846	0.756	0.776	0.877	0.828
ABIDE / Sex	0.521	0.514	0.492	0.694	0.538
ADNI / NC-MCI	0.567	0.497	0.484	0.566	0.604
ADNI / AD-HC	0.481	0.537	0.577	0.591	0.731

Axis	base	fourier	noblock2	pool	unfrozen
ADNI / Amyloid	0.611	0.539	0.500	0.641	0.640
HCP / Sex	0.908	0.900	0.933	0.962	0.886
HCP / Age	0.633	0.594	0.626	0.680	0.592
OASIS / AD Conv	—	—	—	—	—

Metric: test AUROC (linear probe, 30% held-out). OASIS = labels missing.

4. Probe ablations (on base)

4a. Temporal aggregation of the CLS token — AUROC

Axis	mean (384-d)	mean_std (768-d)	delta
ABIDE / Autism	0.605	0.583	-0.023
ABIDE / Age	0.846	0.835	-0.011
ADNI / NC_vs_MCI	0.567	0.505	-0.062
ADNI / Amyloid	0.611	0.606	-0.005
ADNI / AD_vs_HC	0.481	0.404	-0.077
HCP / Sex	0.908	0.853	-0.055
HCP / Age	0.633	0.595	-0.037
ADHD / ADHD	0.552	0.557	+0.005
COBRE / Schizophrenia	0.519	0.538	+0.019

*mean_std mainly helps task dynamics; elsewhere **mean** wins.*

4b. MLP head architecture — AUROC

Axis	128	256	256x128	512x256	512x256x128	linear
ADNI / NC_vs_MCI	0.588	0.509	0.523	0.475	0.433	0.567
ADNI / AD_vs_HC	0.407	0.467	0.472	0.448	0.487	0.481
ADNI / Amyloid	0.621	0.623	0.610	0.631	0.610	0.611
ABIDE / Autism	0.550	0.497	0.561	0.509	0.522	0.605
HCP / Sex	0.844	0.906	0.896	0.898	0.895	0.908
ADHD / ADHD	0.477	0.466	0.382	0.412	0.397	0.552
COBRE / Schizophrenia	0.602	0.566	0.595	0.557	0.590	0.519

The MLP head does not clearly beat the linear probe; deeper heads overfit.

5. Best results vs SOTA (same-dataset comparisons)

Caveats: (1) Brain-JEPA numbers are **fine-tuning**, ours are **linear probe**; (2) we report acc/F1 of the **best-AUROC** config (not the raw max accuracy, which would reward a majority-class classifier).

Brain-JEPA (same dataset = ADNI)

Benchmark	Metric	Ours (best-AUC)	config	Brain-JEPA (FT)
ADNI / NC-MCI	Acc	0.576	unfrozen_adni	0.768
ADNI / NC-MCI	F1	0.632	unfrozen_adni	0.863
ADNI / Amyloid	Acc	0.559	pool_adni	0.710
ADNI / Amyloid	F1	0.623	pool_adni	0.760

NeuroSTORM (same dataset = ADHD-200)

Benchmark	Metric	Ours (best-AUC)	config	NeuroSTORM
ADHD-200	Acc	0.617	base_adhd_agg-mean_std	0.587
ADHD-200	AUROC	0.557	base_adhd_agg-mean_std	—

Not compared (different dataset): HCP Sex/Age (HCP-YA vs HCP-Aging), COBRE vs HCP-EP, UCLA (to download). OASIS/ABIDE are not Brain-JEPA benchmarks.

SOTA sources: Brain-JEPA (arXiv 2409.19407, Tables 2-3, fine-tuning) · NeuroSTORM (arXiv 2506.11167).